Backtesting Trading Strategies Across Bullish ETFs and a Bearish Stock

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Methodology and Assumptions

The trading strategy backtest uses five years of historical daily OHLCV data from Alpaca, covering July 2, 2021 to June 30, 2026. The analysis compares a buy-and-hold benchmark against three long-only rule-based strategies. Each strategy is either fully invested in the selected asset or fully in cash, does not use short selling, margin, or leverage. Strategy returns are calculated using daily close-to-close percentage returns, with positions applied after the trading signal so there is no intraday trading.

The backtest assumes an initial portfolio value of $100,000 for each asset and strategy. Transaction costs, bid-ask spreads, slippage, taxes, funding costs, and dividend reinvestment are not included, so the reported results may be higher than actual live trading performance. The data is based on the daily bars returned by Alpaca, and no separate price adjustment step is applied in our code. Performance is evaluated using CAGR, annualized volatility, Sharpe ratio, and maximum drawdown. The Sharpe ratio is calculated as the average strategy return divided by the standard deviation of strategy returns, annualized using 252 trading days, with no explicit risk-free rate adjustment (assuming US treasury bill rate of 0%).

Strategy Descriptions and Rules

Strategy 1: Trend Following

Trend-following strategies are based on the idea that assets moving strongly in one direction are likely to continue in that direction for some time. These strategies use technical indicators to identify the strength and direction of a trend, helping determine when to enter or exit a position. The strategy is designed to participate in an upward trend and move to cash when momentum weakens. Our strategy combines a Moving Average Convergence Divergence (MACD) indicator with a 200-day Simple Moving Average (SMA) filter. The rationale behind the strategy is to rely on the MACD indicator to capture momentum changes, while the SMA 200 helps avoid long exposure when the asset is below its long-term trend.

Entry rule:
MACD > MACD signal && Close > SMA-200

Exit rule:
MACD < MACD signal || Close < SMA-200

Strategy 2: Mean Reversion

Mean reversion strategies are based on the idea that prices often move too far in one direction in the short term and eventually return closer to their normal range. Instead of trying to follow a strong trend, this strategy looks for moments when the asset appears temporarily overextended. When the price appears unusually low relative to its recent behavior, the strategy treats it as a potential buying opportunity. When the price moves unusually high, the strategy exits the position to lock in gains or avoid a reversal. Overall, the goal of the strategy is to take advantage of short-term price corrections rather than long-term directional moves. Our strategy combines a Relative Strength Index (RSI) indicator with Bollinger Bands (BBs). The rationale behind this strategy is to look for oversold conditions confirmed by a volatility-adjusted price band, then exit after a strong rebound.

Entry rule:
RSI-14 < 30 && Close < lower BB-20

Exit rule:
RSI-14 > 70 && Close > upper BB-20

Strategy 3: Custom Multi-Factor Strategy

Our custom multi-factor strategy combines indicators from three categories: trend, momentum, and volatility. The goal is to enter only when multiple independent signals point to a strong upside trend. The trend indicators used for this strategy are SMA 50, SMA 200, and EMA 20; the momentum indicators are MACD and RSI; and the volatility indicators are Bollinger Bands. The rationale behind the strategy is that the trend filter avoids weak long-term environments, the momentum filter reduces false breakouts, and the Bolling Band breakout requires evidence of strong volatility expansion.

Entry rule:
Close > SMA-200 && EMA-20 > SMA-50 && MACD > MACD signal && 50 < RSI-14 < 70 && Close > upper BB-20

Exit rule:
Close < EMA 20 || MACD < MACD signal || RSI-14 < 45 || Close < middle BB-20

Performance Overview

We tested the strategies on Vanguard S&P 500 ETF (VOO), Invesco QQQ Trust (QQQ), and PayPal Holdings, Inc. (PYPL) over the five-year backtest period, from July 2, 2021 to Jun 30 2026.. VOO and QQQ are included as bullish equity examples, while PYPL is included as a bearish equity example. The backtest results are reported with metrics including CAGR, volatility, Sharpe ratio, and maximum drawdown for the buy-and-hold benchmark and each trading strategy.

Vanguard S&P 500 ETF **(VOO)**

Strategy	CAGR	Volatility	Sharpe Ratio	Max Drawdown
Buy-and-hold	11.70%	16.86%	0.74	-25.32%
Strategy 1: Trend Following	3.63%	5.90%	0.63	-5.87%
Strategy 2: Mean Reversion	7.04%	14.83%	0.53	-24.70%
Strategy 3: Multi-Factor	-0.74%	2.74%	-0.26	-5.43%

Table 1. Five-year backtest performance for Vanguard S&P 500 ETF (VOO) using daily candlestick OHLCV. The table compares the buy-and-hold benchmark with three rule-based trading strategies across CAGR, volatility, Sharpe ratio, and maximum drawdown.

Invesco QQQ Trust **(QQQ)**

Strategy	CAGR	Volatility	Sharpe Ratio	Max Drawdown %
Buy-and-hold	15.80%	22.67%	0.76	-35.63%
Strategy 1: Trend Following	5.37%	8.92%	0.63	-14.86%
Strategy 2: Mean Reversion	6.98%	19.13%	0.45	-34.25%
Strategy 3: Multi-Factor	-4.14%	4.29%	-0.18	-9.95%

Table 2. Five-year backtest performance for Invesco QQQ Trust ETF (QQQ) using daily candlestick OHLCV. The table compares the buy-and-hold benchmark with three rule-based trading strategies across CAGR, volatility, Sharpe ratio, and maximum drawdown.

PayPal Holdings, Inc. Common Stock **(PYPL)**

Strategy	CAGR	Volatility	Sharpe Ratio	Max Drawdown %
Buy-and-hold	-31.76%	42.20%	-0.69	-87.33%
Strategy 1: Trend Following	-10.17%	11.00%	-0.92	-41.35%
Strategy 2: Mean Reversion	-22.37%	36.18%	-0.51	-79.16%
Strategy 3: Multi-Factor	-5.99%	6.72%	-0.89	-26.48%

Table 3. Five-year backtest performance for PayPal Holdings, Inc. (PYPL) using daily candlestick OHLCV. The table compares the buy-and-hold benchmark with three rule-based trading strategies across CAGR, volatility, Sharpe ratio, and maximum drawdown.

Discussion

The backtest results show that the three rule-based strategies mainly reduced risk, but they did not consistently outperform buy-and-hold in terms of return given the overall bullish market in the past 5 years.

For VOO and QQQ, buy-and-hold produced the highest CAGR, with 11.70% for VOO and 15.80% for QQQ. However, this came with higher volatility and larger drawdowns. QQQ, for example, had a maximum drawdown of -35.63%, showing that a fully invested portfolio is subject to larger losses during market declines. However, for PYPL, which declined heavily driven by weaker post-pandemic growth and stronger competition in digital payments compared with its earlier high-growth valuation, buy-and-hold performed very poorly with a maximum drawdown of -87.33%.

Strategy 1, the trend-following strategy, performed best as a defensive strategy. It had lower CAGR than buy-and-hold, but it sharply reduced volatility and drawdowns. For VOO, the maximum drawdown fell from -25.32% under buy-and-hold to only -5.87%. For QQQ, it fell from -35.63% to -14.86%. This makes sense because the strategy only enters when momentum confirms an upward trend, so it avoids the weak market periods. The apparent tradeoff is that it can enter late and miss part of a rebound. The trend following strategy still lost money in the bearish PYPL with a -10.17% CAGR and -41.35% drawdown, but it was a much smaller loss nearly by half compared to the benchmark.

Strategy 2, the mean-reversion strategy, produced moderate results for VOO and QQQ that were slightly better than Strategy 1, but it struggled more on PYPL. It assumes that short-term weakness will reverse, which worked better for the two bullish ETFs than for a single stock in a persistent decline. PYPL's mean-reversion strategy had a -22.37% CAGR and a -79.16% maximum drawdown, showing that buying dips and expecting future correction can further accumulate portfolio loss when the asset continues falling.

Strategy 3, the multi-factor strategy, was the most conservative among the three. It had the lowest volatility and smallest drawdowns across the assets, including only 6.72% volatility and a -26.48% drawdown for PYPL. The trade-off was its negative CAGR for VOO and QQQ in a strong bullish historical period. The enter and exit rules were too restrictive and kept the strategy out of profitable periods.

Overall, the results suggest that buy-and-hold was most effective when the underlying asset had a strong long-term upward trend, especially for broad ETFs like VOO and QQQ. However, the rule-based strategies were useful for managing downside risk because they reduced volatility and avoided larger drawdowns. The PYPL results highlight this tradeoff clearly: when an individual stock experiences a sustained negative trend, risk control becomes more important than simply staying invested.

Our backtest has several limitations. First, the results do not include transaction costs, bid-ask spreads, stock-split, or slippage, so the actual trading performance may be lower than reported, especially for strategies that trade more frequently. Second, the five-year sample period may not represent all market environments, since VOO and QQQ were generally bullish while PYPL was strongly bearish. Finally, the selected assets were chosen as examples of bullish ETFs and a bearish individual stock, which may introduce selection bias. Therefore, the backtest should be interpreted as a comparison of strategy behavior under specific market conditions, not as proof of future performance.